DISPROOF OF EVENTUAL EQUALITY IN ERDŐS PROBLEM 878 (QUESTION 3B)

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For an integer $n \geq 1$, let $\mathcal{P}(n)$ denote the set of prime factors of n, let $\omega(n) := |\mathcal{P}(n)|$, and for a prime p let $q_p(n) := p^{\lfloor \log_p n \rfloor}$ be the largest power of p not exceeding n. Erdős [1] defined

$$f(n) := \sum_{p \in \mathcal{P}(n)} q_p(n), \qquad F(n) := \max \sum_{a \in A} a,$$

where $A \subseteq [2, n]$ ranges over finite sets of pairwise coprime integers a with $\mathcal{P}(a) \subseteq \mathcal{P}(n)$, and asked whether the maxima $M_f(x) := \max_{n \leq x} f(n)$ and $M_F(x) := \max_{n \leq x} F(n)$ satisfy $M_f(x) = M_F(x)$ for all sufficiently large x.

Theorem. *For every $X_0 \in \mathbb{N}$, there exists $x \geq X_0$ such that $M_f(x) < M_F(x)$.*

Lemma 1. *If $p < r$ are primes with $98r < 100p$, then for every $n \geq p^{p^2+2}$ there exists an integer $z \in [2, n]$ with $\mathcal{P}(z) \subseteq \{p, r\}$ and $98n < 100z$.*

Proof. Put $\ell = \lfloor \log_p n \rfloor$, $J = p^2 + 1 < \ell$, and $z_j = p^{\ell - j} r^j$. Then $z_0 = q_p(n) \leq n$, while

$$z_J \geq p^{\ell - J}(p+1)^J \geq p^{\ell - J}(p^J + Jp^{J-1}) = p^{\ell - J}\big(p^J + (p^2+1)p^{J-1}\big) > p^{\ell+1} > n.$$

Choose $0 \leq j < J$ with $z_j \leq n < z_{j+1}$. Since $z_{j+1}p = z_j r$, we have $98np < 98 z_{j+1} p = 98 z_j r < 100 z_j p$, so $z = z_j$ has the required properties. □

Fix the six primes

$$Q_6 := \{168000061,\ 168000101,\ 168000121,\ 168000139,\ 168000143,\ 168000167\}$$

and let $C_6 := \prod_{q \in Q_6} q$. Applying Lemma 1 to the pairs (q_1, q_2), (q_3, q_4), and (q_5, q_6) yields a threshold N_0 such that for every $m \geq N_0$ there exist $b_1, b_2, b_3 \in [2, m]$ with $\mathcal{P}(b_i) \subseteq \{q_{2i-1}, q_{2i}\}$ and $100 b_i > 98m$. Whenever $C_6 \mid m$ and $R' \subseteq \mathcal{P}(m) \setminus Q_6$, the integers b_1, b_2, b_3 and $\{q_p(m)\}_{p \in R'}$ are pairwise coprime with prime factors dividing m, so

$$100 F(m) > 294m + 100 \sum_{p \in R'} q_p(m). \tag{1}$$

Lemma 2. *For all $n \geq 1$, $f(n) < f(2n)$. Consequently, if $f(n) = M_f(y)$ with $n \leq y$, then $n > y/2$.*

Proof. Passing from n to $2n$ never decreases q_p for $p \in \mathcal{P}(n)$, and strictly increases the $p = 2$ term (either $q_2(2n) = 2q_2(n)$ if $2 \mid n$, or $q_2(2n) \geq 2$ is adjoined if $2 \nmid n$). □

Lemma 3. *For every $R > 0$, $M_f(y) > Ry$ for all sufficiently large y.*

Proof. Since $\sum_p 1/p = \infty$, choose N with $\sum_{p<N} 1/p \geq 2R + 1$ and set $P = \prod_{p<N} p$. For $y \geq 2P$, let $m = P\lfloor y/P \rfloor > y/2$. Every prime $p < N$ divides m and satisfies $q_p(m) > m/p$, so

$$M_f(y) \geq f(m) \geq \sum_{p<N} q_p(m) \geq m \sum_{p<N} \frac{1}{p} \geq (2R+1)m > Ry. \qquad \square$$

Lemma 4. *For every X_0, there is $Y \geq \max(X_0, 20)$ with $14 q_p(\lfloor 21Y/20 \rfloor) < Y$ for all $p \in Q_6$.*

Proof. Let $Y_m = \max(X_0, 20) \cdot 15^m$ for $0 \leq m \leq 6$. If a prime $p \in Q_6$ fails the inequality at two indices $m_1 < m_2$, then $Q_a := q_p(\lfloor 21 Y_{m_1}/20 \rfloor) \geq Y_{m_1}/14$ and $Q_b := q_p(\lfloor 21 Y_{m_2}/20 \rfloor) \geq Y_{m_2}/14 \geq \frac{15}{14} Y_{m_1} > Q_a$. Since Q_a, Q_b are powers of p,

$$\frac{21}{20} Y_{m_2} \geq Q_b \geq p Q_a \geq \frac{168000000}{14} Y_{m_1} = 12000000 Y_{m_1},$$

contradicting $Y_{m_2} \le 15^6 Y_{m_1} = 11390625 Y_{m_1}$. Thus each of the 6 primes in Q_6 eliminates at most one of the 7 exponents $m \in \{0, \dots, 6\}$. □

Proof of the Theorem. Suppose for contradiction that $M_F(x) \le M_f(x)$ for all $x \ge X_0$.

Step 1. Let $Y \ge \max(X_0, 20, 2N_0)$ satisfy Lemma 4. Suppose $y \in [Y, \lfloor 21Y/20 \rfloor]$ and $M \ge 1$ satisfy $100M_f(y) \ge 105(M+8)Y$. Choose $n \le y$ with $f(n) = M_f(y)$.

- By Lemma 2, $n > y/2 \ge N_0$.
- Since $f(n) \le \omega(n)n$ and $20n \le 21Y$, we have $105(M+8)Y \le 100\omega(n)n \le 105\omega(n)Y$, so $\omega(n) \ge M+8$.
- Removing Q_6 (≤ 6 primes) and primes $< M$ ($\le M$ integers) from $\mathcal{P}(n)$ leaves at least two distinct primes $r_a, r_b \ge M$ outside Q_6.
- Put $R' := (\mathcal{P}(n) \setminus Q_6) \setminus \{r_a, r_b\}$ and $D := C_6 \prod_{p \in R'} p$. Since $r_a r_b \prod_{p \in R'} p \mid n \le 2Y$, we have
$$D \le \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor.$$
- Let $m := D(\lfloor n/D \rfloor + 1) \in (n, y+D]$. By Lemma 4, each $q \in Q_6$ satisfies $14q_q(n) \le 14q_q(\lfloor 21Y/20 \rfloor) < Y < 2n$, so $\sum_{q \in Q_6} q_q(n) < \frac{6}{7}n$. Using $q_{r_a}(n), q_{r_b}(n) \le n$ and $q_p(n) \le q_p(m)$ for $p \in R'$,
$$f(n) < 2n + \frac{6}{7}n + \sum_{p \in R'} q_p(n) \le \frac{20}{7}n + \sum_{p \in R'} q_p(m).$$
Comparing with (1) gives $F(m) - f(n) > \left(\frac{294}{100} - \frac{20}{7}\right) n = \frac{29}{350}n > \frac{29}{700}y \ge \frac{29}{700}Y > \frac{Y}{25}$. Since $m \le y + D$ and $M_F(y+D) \le M_f(y+D)$, setting $y' = y + D$ yields

$$y' \le y + \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor \quad \text{and} \quad 25M_f(y') \ge 25M_f(y) + Y. \tag{2}$$

Step 2. Set $A := 2000C_6 + 1$ and $R_k := \frac{54C_6 Y}{A+k}$. By Lemma 3, we may choose Y above so that $M_f(Y) > 2(A+10)Y$. We construct $y_k \ge Y$ by induction on $k \ge 0$ with
$$20(y_k + R_k) \le 21Y \quad \text{and} \quad 100M_f(y_k) \ge 105(A+9+k)Y.$$
For $k = 0$, $y_0 = Y$ satisfies both inequalities since $20R_0 = \frac{1080C_6 Y}{2000C_6+1} < Y$. Given y_k, set $M = A + 1 + k$. Over 27 steps of size $\lfloor 2C_6 Y/M^2 \rfloor$, the total increase in y is at most
$$27 \left\lfloor \frac{2C_6 Y}{(A+1+k)^2} \right\rfloor \le \frac{54C_6 Y}{(A+k)(A+k+1)} = R_k - R_{k+1}.$$
Since $R_{k+1} > 0$, all 27 intermediate values remain $\le \lfloor 21Y/20 \rfloor$, so applying (2) 27 times gives y_{k+1} with $20(y_{k+1} + R_{k+1}) \le 21Y$ and $100M_f(y_{k+1}) \ge 100M_f(y_k) + 27 \cdot 4Y = 100M_f(y_k) + 108Y > 105(A+10+k)Y$.

Taking $k = M_f(\lfloor 21Y/20 \rfloor) + 1$ gives $y_k \le \lfloor 21Y/20 \rfloor$ and $M_f(y_k) \ge k > M_f(\lfloor 21Y/20 \rfloor)$, a contradiction. □

References

[1] P. Erdős, *On two unconventional number theoretic functions and on some related problems*, Topics in classical number theory, Vol. I, II (Budapest, 1981), Colloq. Math. Soc. János Bolyai **34** (1984), 445–458.